

11 useful readings, then we choose the next step

Start with the tuning controls, receiver reference/supplies, and the repeated capacitor cells.

1 Set the contact baseline

Disconnect battery and programmer. Short the probes, then measure the known V022-V016 connection once. Write both readings. Use actual ohms, kohms, OL, or changing; a beep alone can include a resistor path.

Shorted: _____ ohm

V022-V016: _____ ohm

2 Measure this first batch

Keep probe A fixed within each group. Stop A at the first verified direct hit; complete all B and C rows.

Test	Probe A	Probe B	Why this pair	Reading	Reversed if low
A1	V047	V108	RC0 to R14: possible tuning control	_____	_____
A2	V047	V102	RC0 to R15: next tuning candidate	_____	_____
A3	V047	V097	RC0 to R16: next tuning candidate	_____	_____
A4	V047	V091	RC0 to R11: next tuning candidate	_____	_____
B1	V053	V071	VREF-labelled pad to R21 receiver bias	_____	_____
B2	V064	V063	Regulated VDD to R37/R39 feed	_____	_____
B3	V064	V070	Regulated VDD to R18 receiver feed	_____	_____
C1	V090	V095	Capacitor bank: C46/C47 to C43/C44	_____	_____
C2	V090	V100	Capacitor bank: C46/C47 to C16/C29	_____	_____
C3	V090	V103	Capacitor bank: C46/C47 to C13/C14	_____	_____
C4	V090	V107	Capacitor bank: C46/C47 to C10/C11	_____	_____

3 Interpret, without turning guesses into wires

A direct hit should be stable and close to the connected-pair baseline, with probes reversed as a cross-check. Tens/hundreds of ohms, polarity-dependent readings or charging behaviour do not prove direct copper. Even near-zero may include a zero-ohm link or an inductor.

A is a candidate sweep: we know RC6 reaches R13, but do not know the other four assignments. Do not infer the last pair by elimination without measuring it.

After these results

We will test RC2 and accessible TP11/TP17 against the unmatched tuning controls; choose supply/reference alternatives only when needed; and test one representative of each confirmed capacitor group to ANT1/ANT2. V011-V015 is already excluded. DNP V067 can wait.

Photo locator: docs/VIA_PAIR_TESTS.html. Two zoomed points per test, with a copyable reply form. Photo atlas on page 2.

Where to put the probes

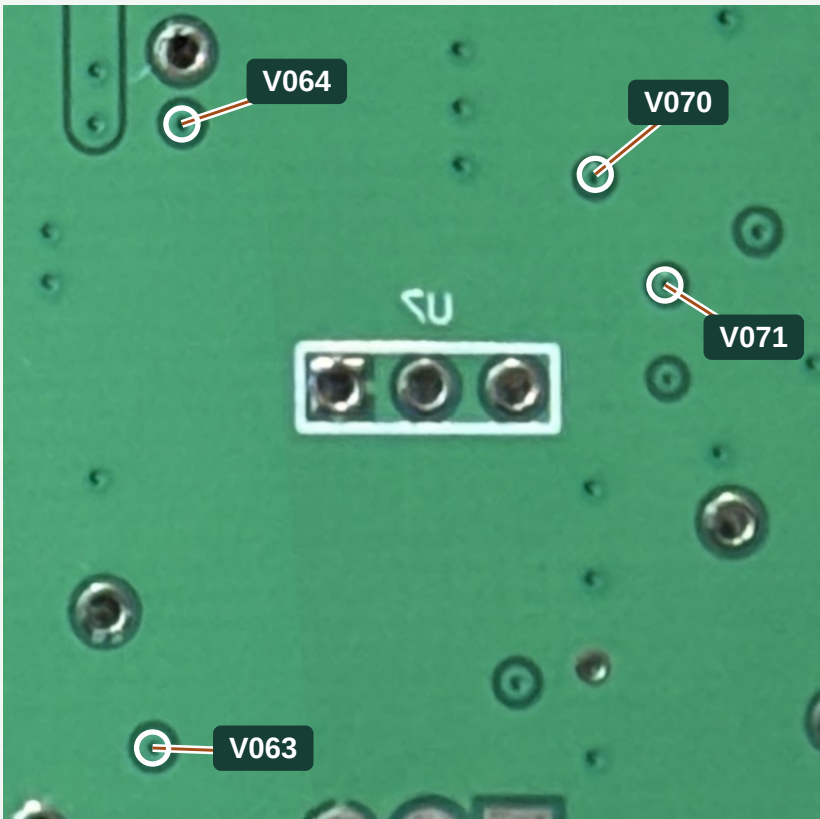
Rear mosaic is mirrored to match the front view. Circles identify sites; leaders are annotations, not traces.

A anchor / B1 reference



V047 = PIC pin11 / RC0. V053 = PIC pin12 / VREF pad.

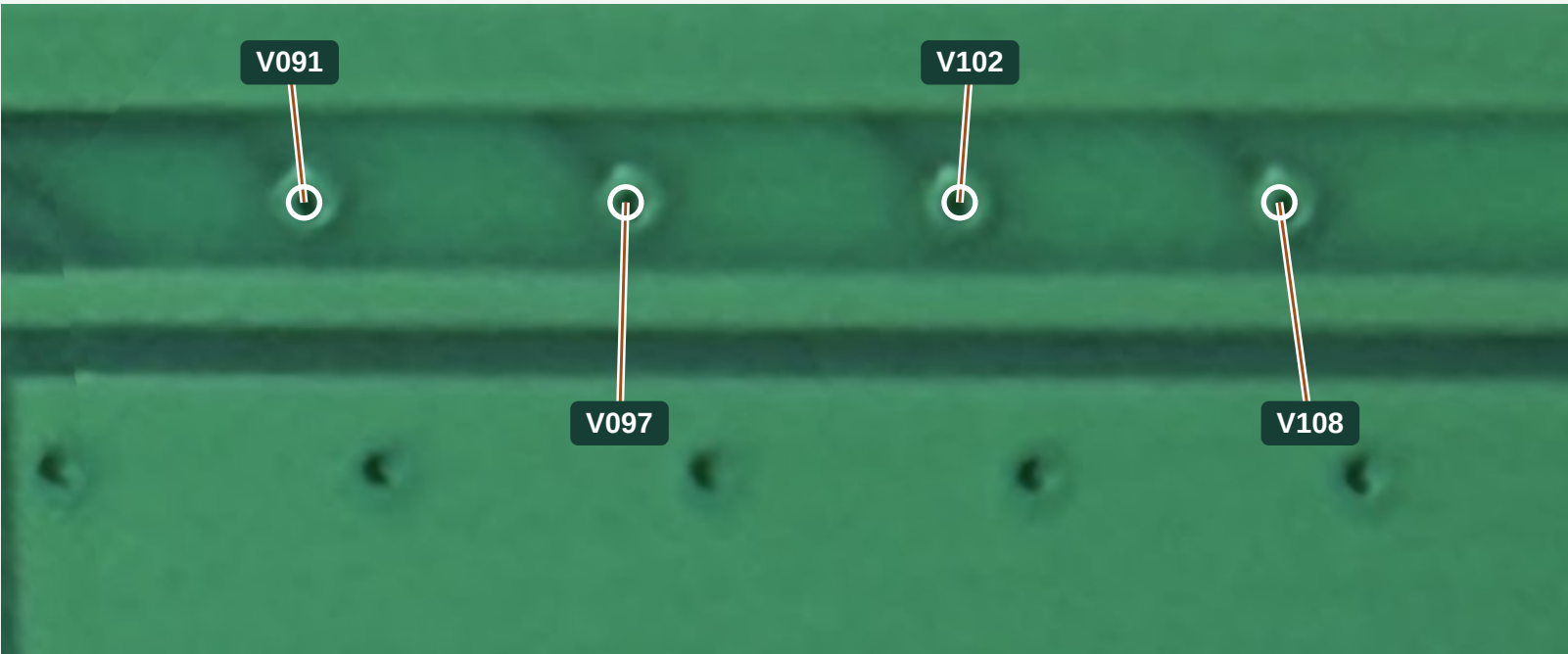
B reference / supply candidates



B1: V053 - V071 | B2: V064 - V063
| B3: V064 - V070

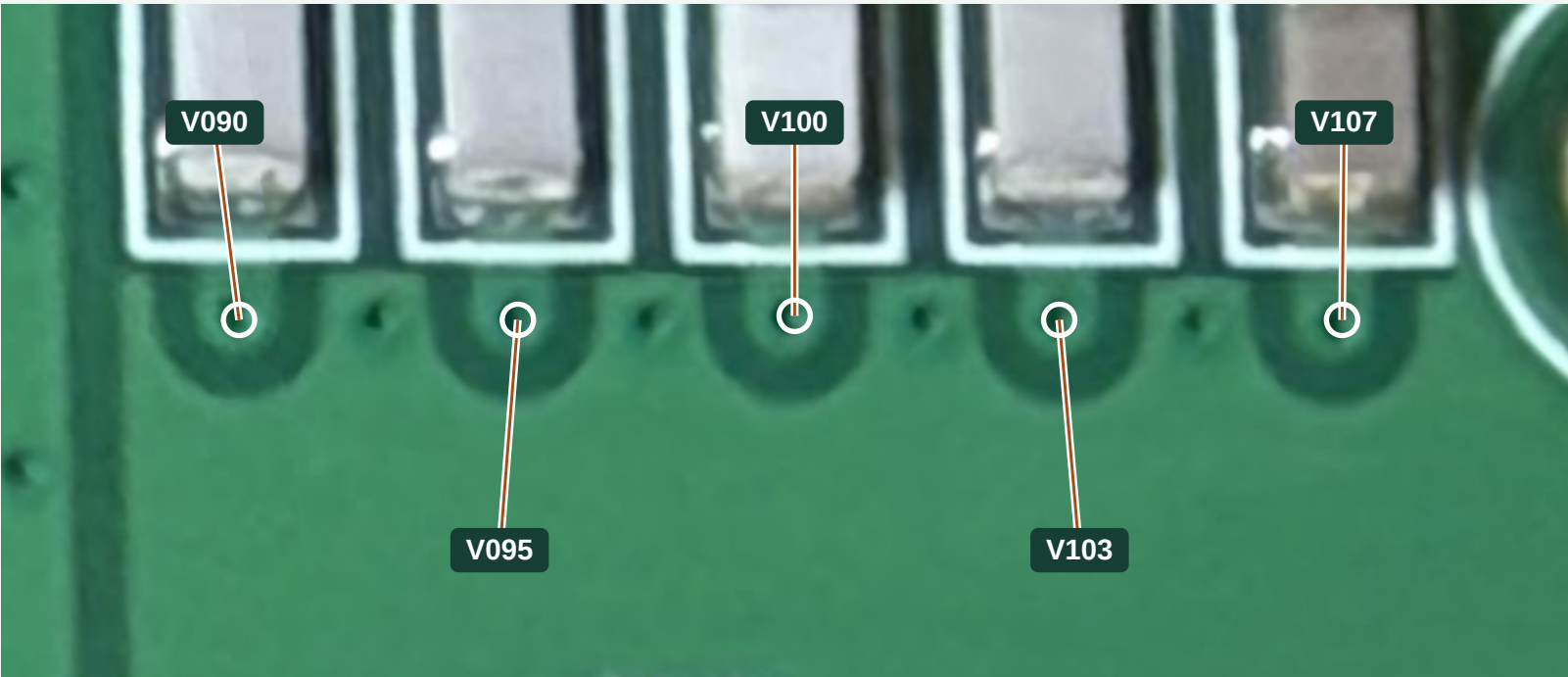
V064 is regulated VDD.
Motor/battery VDD is a separate
group; no rail merge is assumed.

A moving probe: four resistor-control vias



Right-to-left sweep: V108/R14, V102/R15, V097/R16, V091/R11.

C capacitor-bank junctions: hold V090



Only labelled junctions: the adjacent rear capacitor pads include GND.